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A Multifaceted Approach to Bangla Speech Emotion Recognition Using Hybrid Feature Fusion and Dimensionality Reduction

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***Abstract*—Speech Emotion Recognition (SER) is critical for Human-Computer Interaction (HCI), but robust SER systems for the Bangla language remain underexplored. This paper proposes a novel approach that combines extensive fusion of acoustic characteristics with statistical descriptors and a multifaceted approach based on dimensionality reduction to improve Bangla SER performance. We fuse diverse time-domain, frequency-domain, and time-frequency features (e.g., Amplitude Envelope, Zero-Crossing Rate, Mel-frequency cepstral coefficients, Spectral Centroid) by computing summary statistics. We apply Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Linear Discriminant Analysis (LDA) to create a compact multifaced feature set. The approach is evaluated on two Bangla emotional speech datasets—SUBESCO (7 emotions) and KBES (9 emotions)—using four machine learning classifiers (k-Nearest Neighbors, Decision Tree, Random Forest, and XGBoost). The best model, Random Forest with the proposed feature-reduction pipeline, achieves 97.78% accuracy on the KBES dataset and 87.14% on the larger SUBESCO dataset using only a fraction of the original features. These results demonstrate that the proposed strategy can significantly enhance emotion recognition accuracy while greatly decreasing feature dimensionality. This improvement will increase the efficiency of SER systems for under-resourced languages, such as Bangla.**

***Index Terms*—Speech Emotion Recognition, Feature Fusion, Dimensionality Reduction, Multifaceted Dataset, Acoustic Features, Human-Computer Interaction.**

I. Introduction

Speech is one of the most important means to express human emotions [1]. Emotions can reflect human psychological conditions (e.g. mood, anxiety, trauma) [2]. After the COVID-19 pandemic, the education system and medical consultations (e.g. mental health hotlines) are increasingly shifting to online platforms [3] [4] [5]. Nowadays, it is essential to teach computers to detect human emotions automatically. Using a Speech Emotion Recognition (SER) system, it is possible to detect students' reactions to their tutors and also assess patients' mental health conditions [6] [7]. SER is a part of Human-Computer Interaction (HCI), which plays a significant role in modern computer games, robotics, etc. [8].

The acoustic features of speech play a crucial role in emotion recognition. Time-domain audio features include Amplitude Envelope, Root-Mean-Square Energy (RMS), and Zero Crossing Rate (ZCR) [9] [10] [11]. Frequency-domain audio features include Band Energy Ratio (BER), Spectral Centroid [12], and Spectral Flux. Finally, Time-frequency domain features include spectrograms [13], mel-spectrograms and Mel-Frequency Cepstral Coefficients (MFCCs) [9] [14] [12] [10] [13]. These features are essential for developing robust Machine Learning (ML) models that accurately classify emotions. Data reduction techniques such as Principal Component Analysis (PCA) [15], Independent Component Analysis (ICA) [15], and Linear Discriminant Analysis (LDA) [10] are also used in SER systems. While SER systems are well-developed for languages such as English and German, they are still comparatively underdeveloped for Bangla. In our study, we focus on establishing Bangla SER systems.

We propose a Bangla SER model that fuses audio features using discrete statistical measures (mean, median, minimum, maximum, and standard deviation) and applies data reduction techniques to enhance feature extraction and selection. The audio features are extracted using Amplitude Envelope, RMS, ZCR, Mel-Spectrogram, MFCC, BER, Spectral Centroid, and Spectral Bandwidth. We then apply PCA, ICA, and LDA techniques individually for dimensionality reduction and combine all the resulting components to introduce a new Multifaceted dataset [15] [16]. Then classified with machine learning models, including K-Nearest Neighbors, Decision Tree, Random Forest, and XGBoost, to assess recognition accuracy across various fusion strategies. In this study, two datasets are investigated for the Bangla language: the SUST Bangla Emotional Speech Corpus (SUBESCO) [17] and the KUET Bangla Emotional Speech (KBES) [18], both developed in recent years and made publicly available. Key contributions of this study are:

- We analyze the correlation between feature selection and dimensionality reduction and their impact on the performance of the SER models.
- We introduce a unique multifaceted dataset by using a reduction approach to identify SER's most effective

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feature combinations.

- We evaluate multiple machine learning classifiers along with hyperparameter tuning to determine the best-performing SER model.

Our research aims to improve the accuracy and efficiency of SER, particularly in underexplored languages like Bangla. By leveraging advanced feature selection and dimensionality reduction techniques, this work contributes to the growing body of knowledge in SER. It will open new avenues for its application in real-world scenarios.

II. Literature Review

SER is an active research area. Many researchers are working to develop SER models using Machine Learning (ML) and Deep Learning (DL) models. One of the studies is conducted by Esteban Garcia-Cuesta *et al.* (2024) entirely in Spanish. They used the EMOSpanish DB and introduced a new dataset, EMOMatchSpanishDB, with 2050 audio files. Despite using expanded features such as MFCC, Spectral Centroid, Spectral Flatness, Spectral Contrast, and Linear Predictive Coefficients (LPC), the SVM model achieved only 65% accuracy in classifying 7 emotions across both datasets [12]. Similarly, Rafael Kingeski *et al.* (2024) applied PCA and ICA for dimensionality reduction in SER models for German and English languages. They reduced 6373 features to 170, 130, and 150 for the EMO-DB, SAVEE, and RAVDESS datasets, respectively. Using the SVM classifier, they achieved 84.3%, 75.4%, and 59.9% accuracy for these datasets [15].

Other researchers focused on DL models as well. Li-Min Zhang *et al.* (2023) conducted two studies, one on English-Chinese and the other on English-German. The first study used RAVDESS and CASA datasets containing 1200 audio files and 5 emotions and applied MFCC, ZCR, Mel, energy, short-time energy, and Chroma features. A Convolutional Neural Network (CNN) model achieved 84.72% and 87.91% accuracy, respectively. In the second study, they introduced a novel feature selection algorithm called F-estimation and used a Multiple Deep Neural Network (multi-DNN) architecture. They achieved 82.3% accuracy on the RAVDESS dataset and 88.8% on the EMO-DB dataset [9] [19]. Similarly, Om Prakash Yadav *et al.* (2021) developed a deep learning model based on a Convolutional Recurrent Neural Network (CRNN) for English and German languages, using the RAVDESS and EMO-DB datasets, respectively. With log-Mel spectrogram features, they achieved 85.76% accuracy on the RAVDESS dataset containing 7356 audio files (classifying 5 emotions) and 91.59% accuracy on the EMO-DB dataset containing 535 audio files (classifying 7 emotions) [20].

Research has also been conducted in the Bangla language, focusing on dataset development and implementing ML and DL models. For example, Sadia Sultana *et al.* (2021) developed one of the largest Bangla SER datasets, SUBESCO, with 7000 audio files and 7 emotions. They also used the English RAVDESS dataset. Using MFCC features and a Deep Convolutional Neural Network (DCNN), they achieved 86.9% accuracy for Bangla SER and 82.7% accuracy for English SER [13]. On the same dataset, Bachchu Paul *et al.* (2024) proposed a multi-language SER model for Bangla, English, and German languages using MFCC, LPC, ZCR, pitch, and energy features. They employed SVM classifiers. They achieved 78.33% accuracy on the SUBESCO dataset [10]. Babe Sultana *et al.* (2025) introduced another Bangla SER system using the BANSpEmo dataset, consisting of 792 audio files and 6 emotions. They used MFCC, spectrogram, ZCR, RMSE, and Chroma features along with SVM, Logistic Regression, and Multinomial Naïve Bayes classifiers. Using SVM, they achieved 87.18% accuracy [11]. Md. Masum Billah *et al.* (2024) utilized the KUET Bangla Emotional Speech (KBES) dataset, which consists of 900 audio files and 9 emotions, and the English RAVDESS dataset to develop an SER model using a novel cascaded deep learning architecture (3D-CNN + TDF + LSTM + Bi-LSTM). They achieved 71.67% accuracy for Bangla and 88.30% for English. [21].

Our study used the publicly available Bangla speech emotion datasets, SUBESCO and KBES, to develop a Bangla SER system. A novel contribution of our work lies in combining three dimensionality reduction techniques: PCA, ICA, and LDA, with feature fusion. This approach enhances classification performance while reducing computational cost, making the system efficient and scalable for Bangla speech emotion recognition using ML methods.

III. Methodology

The overall workflow of the SER system is illustrated in Fig. 1. In the following subsections, we discuss each step extensively.

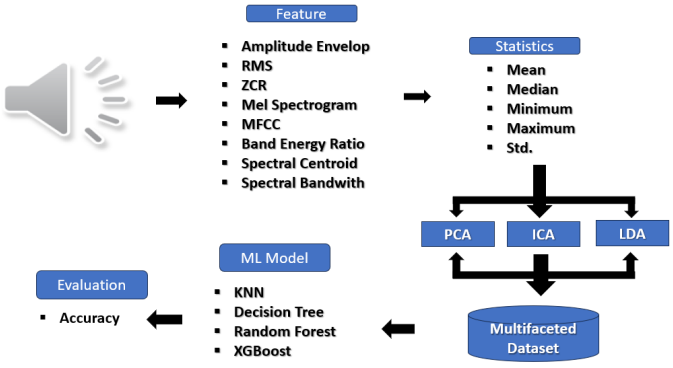


Fig. 1: Methodology of our research

A. Dataset

In this study, we select two popular Bangla speech emotion datasets. One is the SUBESCO Bangla speech dataset with seven emotions, and the other is the KBES Bangla speech dataset with nine emotions. Table I provides detailed descriptions of the two datasets.

B. Data Preprocessing

1) Feature Extraction Techniques: Feature extraction plays a crucial role in SER by transforming raw audio signals into representative feature sets that facilitate accurate emotion

TABLE I: Details of Used Datasets

Attribute	SUBESCO	KBES
Speech Source	Acted in Studio	TV/Film Media
Used Language	Standard Bangla	Standard Bangla
Sampling Rate	48 kHz	48 kHz
Number of Speakers	10 males, 10 females	15 males, 20 females
Age Groups	20–30, 30–40	Not Specified
Number of Emotions	7	9
Emotion States	Neutral, Angry, Disgust, Fear, Happy, Sad, and Surprise	Neutral, Angry, Disgust, Happy, Sad, Very Angry, Very Disgust, Very Happy, Very Sad
Number of Sentences	10	Not Applicable
Total Audio Clips	7000	900

classification. This subsection presents key feature extraction techniques utilized in our study, including their mathematical formulations.

a) Amplitude Envelope: The amplitude envelope represents the signal amplitude variation over time, highlighting the dynamic intensity of speech signals. It is calculated as:

$$AE_t = \max_{k=t.k}^{(t-1)(k-1)} S(k) \tag{1}$$

Where:

- AE_t = Amplitude Envelope at frame t
- k = Sample at time t
- $S(k)$ = Amplitude of the k-th sample

b) Root Mean Square (RMS) Energy: RMS energy provides a measure of the signal's power over time, capturing variations in loudness:

$$RMS_t = \sqrt{\frac{1}{k} \sum_{k=t.k}^{(t-1)(k-1)} S(k)^2} \tag{2}$$

Where, RMS_t = Root Mean Square at frame t

c) Zero Crossing Rate (ZCR): ZCR measures the rate at which the signal changes its sign, providing insight into speech texture:

$$ZCR_t = \frac{1}{2} \sum_{k=t.k}^{(t-1)(k-1)} |\text{sgn}(S(k)) - \text{sgn}(S(k-1))| \tag{3}$$

d) Mel-Spectrogram: The Mel scale is a perceptual scale of pitches judged by listeners to be equal in distance from one another. The conversion from frequency f (Hz) to Mel frequency m_f is given by:

$$m_f = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \tag{4}$$

e) Mel-Frequency Cepstral Coefficients (MFCCs): MFCCs provide a compact representation of the spectral envelope of speech. Forty MFCC features are extracted in this study. The steps are:

- **Pre-emphasis:** Boosts high-frequency energy to preserve speech features:

$$S(n) = X(n) - a \cdot X(n-1) \tag{5}$$

  where $S(n)$ is the output, $X(n)$ is the current sample, and a is a constant in the range $0.95 \leq a \leq 1$.
- **Framing:** Speech is divided into 40 ms frames with 75% overlap to ensure complete feature extraction from each segment.

f) Band Energy Ratio (BER): BER measures the spectral energy ratio in different frequency bands:

$$BER_t = \frac{\sum_{n=1}^{f-1} m_t(n)^2}{\sum_{n=f}^{N} m_t(n)^2} \tag{6}$$

Here, $m_t(n)$ = Weight of the n-th frequency bin at frame t

g) Spectral Centroid: Spectral centroid indicates the "center of mass" of the spectrum:

$$SC_t = \frac{\sum_{n=1}^{N} m_t(n) \cdot n}{\sum_{n=1}^{N} m_t(n)} \tag{7}$$

h) Spectral Bandwidth: Spectral bandwidth quantifies the spread of frequencies around the spectral centroid:

$$BW_t = \frac{\sum_{n=1}^{N} m_t(n) \cdot |n - SC_t|}{\sum_{n=1}^{N} m_t(n)} \tag{8}$$

These feature extraction techniques comprehensively represent speech characteristics, enabling robust emotion classification in our proposed SER framework.

C. Data Reduction

In SER, data reduction techniques help improve model performance by reducing dimensionality while preserving essential information. This section describes PCA, ICA, and LDA.

1) Principal Component Analysis (PCA): PCA is an unsupervised technique that reduces high-dimensional data into a lower-dimensional space while maintaining most of its variance. It finds new orthogonal axes, called principal components (PCs), that maximize data variance. PCA assumes linear relationships and centers on covariance, giving merely uncorrelated independent components [15] [22].

2) Independent Component Analysis (ICA): ICA is an unsupervised dimensionality reduction technique that identifies statistically independent and non-Gaussian variables. ICA facilitates improved data visualization, interpretation, clustering, and classification tasks by identifying the underlying sources of variation in audio signals. Unlike Principal Component Analysis (PCA), ICA components are ranked based on statistical independence rather than variance. Consequently, the order of ICA components is arbitrary, and maximizing variance is not the primary objective of the algorithm [15] [22].

3) Linear Discriminant Analysis (LDA): LDA is a supervised method for dimensionality reduction that aims to maximize class separation while minimizing variation within classes. It is commonly used for feature extraction, as it preserves key discriminative information of different classes. The resulting components are ranked by their ability to differentiate classes, making them suitable for classification or regression tasks [22].

D. Multifaceted Dataset

One of the novel contributions of our work is the creation of multifaceted feature datasets. To achieve this, we generated multiple feature variations combining PCA, ICA, and LDA. Each combination provided a different subset of features, and for identifying the best combinations were evaluated based on their classification accuracy.

E. Classifier

In SER, classification algorithms are crucial for predicting emotions from extracted speech features. To classify emotions, we evaluated four different types of classification algorithms: K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), and XGBoost.

IV. Results and Discussion

This section will discuss our findings and how this study impacts the SER system for the Bangla language. The finalized hyperparameter settings for the four machine learning classifiers are in Table II.

TABLE II: Finalized Hyperparameters Used in Classification

Proposed Model	Hyperparameter	Value
KNN	n_neighbors	7
	metric	`minkowski`
DT	max_depth	6
RF	n_estimators	100
XGBoost	n_estimators	200
	max_depth	6
	learning_rate	0.1
	objective	`multi:softmax`

One of the novel contributions in our study is a combination of feature fusion, summary statistics, and data reduction techniques to enhance the performance of a Bangla SER system. We combined several acoustic features for feature fusion, including Amplitude Envelope, ZCR, MFCC with 40 Mel bands, BER, Spectral Centroid, and Spectral Bandwidth. To further enhance the robustness of the features, we applied statistical operations, like mean, median, minimum, maximum, and standard deviation, to each audio feature. Further, we combined the statistical features and applied PCA, ICA, and LDA for data reduction techniques individually, and finally combined PCA, ICA, and LDA to create a new multifaceted dataset. Our study evaluates various classifiers, like KNN, DT, RF, and XGBoost. However, due to the limitations of this paper, we will only present the classification accuracy of RF, as it is the best-performing model for our Bangla SER. Additional details on classification accuracy and other metrics can be found in the attached link[1]. In Figure 2, the classification accuracies of RF classifiers based on combinations of PCA, ICA, and LDA are presented for the SUBESCO and KBES datasets. The figure illustrates the accuracy comparisons across varying PCA variance and ICA components, while keeping the LDA component constant.

In Figure 2a, the RF classifier achieves an accuracy of 87.14% for the SUBESCO dataset, where the PCA variance is set at 85%, ICA components are 15, and LDA components are 6. Similarly, for KBES datasets, Figure 2b, the RF classifier achieves the highest accuracy of 97.78% using only 40 features, with PCA variance at 60%, ICA components set to 15, and LDA components set to 8.

Table III shows the accuracy of all the ML classifiers considered for this study, both with and without data reduction techniques, including their details.

Moreover, in the Figure 3-4, the accuracy of emotions for all the ML classifiers is illustrated. Figure 3a and 4a represent data without data reduction techniques, and Figures 3b and 4b represent data with data reduction techniques for SUBESCO and KBES datasets, respectively. By analyzing these figures, we concluded that data reduction techniques play an important role in SER systems.

Table IV shows that our proposed technique can play an important role in the SER system. Using our multifaceted dataset based on data reduction techniques, the RF classifier achieved 87.14 and 97. 78%, which is the highest for the SUBESCO and KBES dataset compared to Sadia Sultana et al. [13] and Md. Masum Billah et al. [21] respectively.

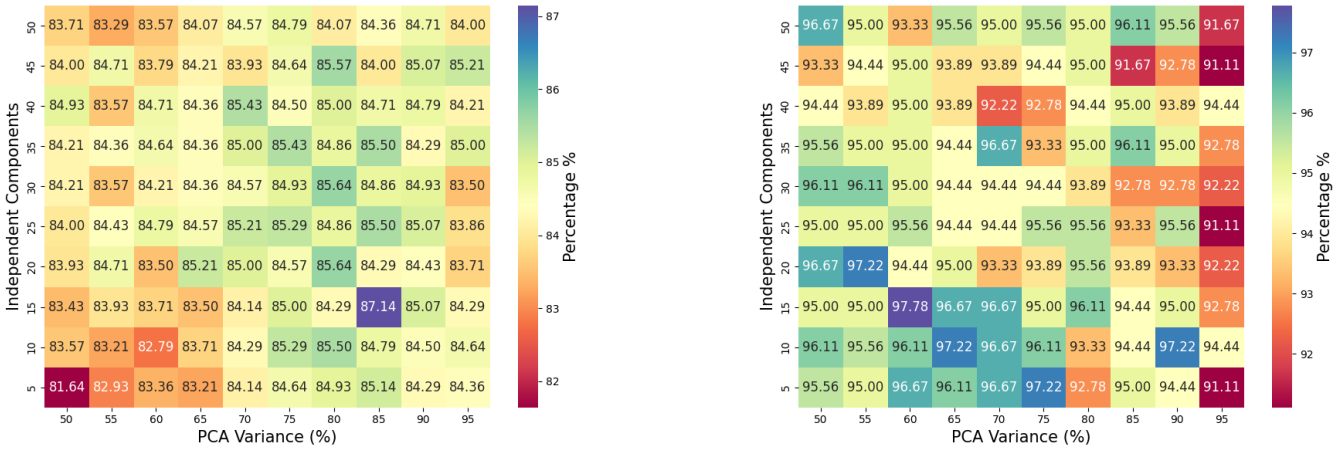
V. Conclusion

This study investigates the efficiency of combining feature fusion, statistical operations, and dimensionality reduction techniques to enhance Bangla SER. We extracted diverse sets of acoustic features using statistical measurements for each feature and fused them to create a comprehensive feature vector capturing the emotional stage in speech. Our data reduction techniques have demonstrated their crucial role in enhancing the accuracy of machine learning classifiers.

With SUBESCO datasets, XGBoost achieved the highest 89% accuracy without data reduction techniques while using 430 features, and RF achieved 87.14% accuracy with data reduction techniques while using only 86 features. With KBES datasets, RF achieved 97.78% accuracy, and XGBoost achieved 93.89% accuracy with data reduction techniques while correspondingly using only 40 and 41 features. RF performs best due to its ensemble property. Our second-best model (XGBoost) has also performed well because of its ensemble property. This reduction not only mitigates the risk of overfitting but also enhances computational efficiency, making the model more practical for real-world applications.

In our future research, we plan to enhance the Bangla SER study by developing a new Bangla speech dataset to benefit

[1]https://github.com/AbuOmayed/Bangla-Speech-Emoiton-Recognition-SER/tree/main

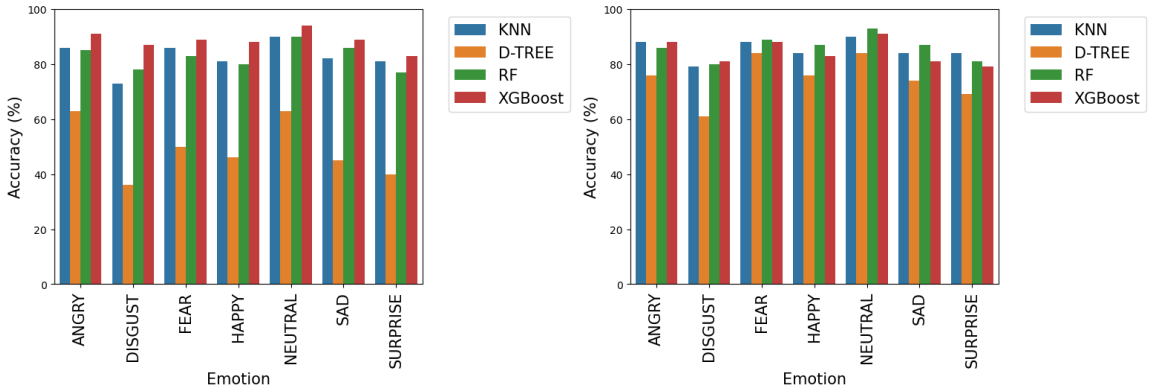


(a) SUBESCO (b) KBES

Fig. 2: Accuracy heatmap of RF for SUBESCO and KBES, PCA variation vs ICA components with LDA constant

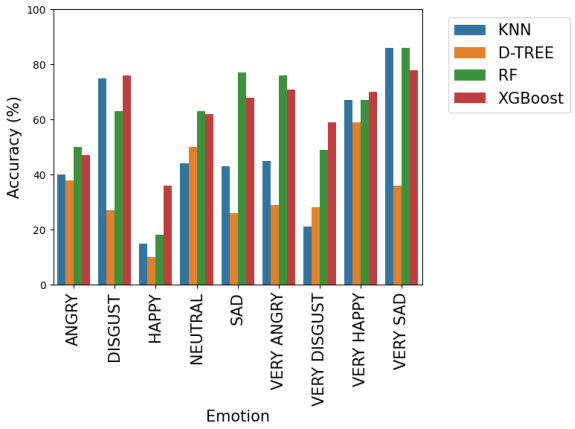
TABLE III: Comparison of Classifier Accuracies Using Feature Fusion and Dimensionality Reduction Techniques

Dataset	Audio Features	Summary Statistics	Reduction Method	Classifier	# Features	Accuracy (%)
SUBESCO	Amplitude Envelope, RMS, ZCR, Mel-Spectrogram, MFCC, Band Energy Ratio, Spectral Centroid, Spectral Bandwidth	Mean, Median, Min, Max, Std.	None	KNN	430	82.24
			None	DT	430	48.13
			None	RF	430	82.93
			None	XGBoost	430	89.06
			PCA+ICA+LDA	KNN	137	85.21
			PCA+ICA+LDA	DT	78	74.93
			PCA+ICA+LDA	RF	86	**87.14**
			PCA+ICA+LDA	XGBoost	43	84.50
KBES	Same as SUBESCO	Mean, Median, Min, Max, Std.	None	KNN	430	50.89
			None	DT	430	34.78
			None	RF	430	59.64
			None	XGBoost	430	63.84
			PCA+ICA+LDA	KNN	23	73.89
			PCA+ICA+LDA	DT	86	80.00
			PCA+ICA+LDA	RF	40	**97.78**
			PCA+ICA+LDA	XGBoost	41	93.89

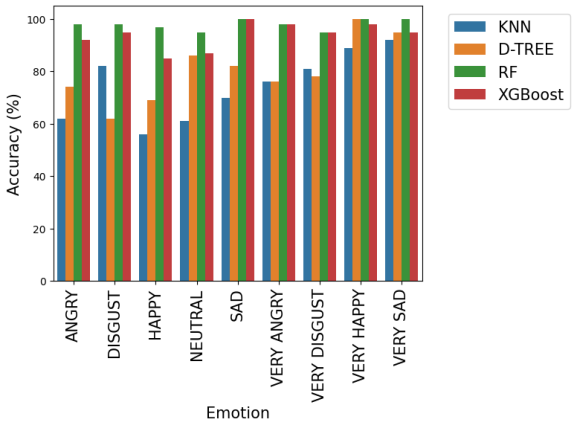


(a) Without Data Reduction (b) With Data Reduction (Multifaceted)

Fig. 3: Emotion recognition Performance by classifiers, (a) Without (b)With, Data Reduction techniques for SUBESCO



(a) Without Data Reduction



(b) With Data Reduction (Multifaceted)

Fig. 4: Emotion recognition Performance by classifiers, (a) Without (b)With, Data Reduction techniques for KBES

TABLE IV: Comparison with Related Works in SER

Author	Year	Dataset	Features	Data Reduction	Best Model	Accuracy
Bachchu Paul et al. [10]	2024	SUBESCO	MFCC, LPC, ZCR, Pitch, Energy	-	SVM	78.33%
Sadia Sultana et al. [13]	2021	SUBESCO	MFCC	-	Deep CNN	86.9%
Md. Masum Billah et al. [21]	2024	KBES	MFCC, STFT, CSTFT	-	Cascaded Deep Learning	71.67%
Proposed Method	-	SUBESCO	Amplitude Envelope, RMS, ZCR, Mel-Spectrogram, MFCC,BER, Spectral Centroid, Spectral Bandwidth	PCA, ICA and LDA	**RF**	**87.14%**
		KBES				**97.78%**

the community. Additionally, we will investigate hybrid deep learning models, such as CNN, LSTM, and GRU, to achieve better emotion recognition and enable real-time processing.

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